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A novel research on a heart disease prediction model based on hybrid intelligent optimization of deep networks

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Abstract
As one of the leading causes of mortality worldwide, the early and precise prediction of heart disease is crucial for effective clinical intervention. This research proposes a novel cardiac prediction framework, which integrates a bidirectional temporal convolutional network (BiTCN) and a bidirectional gated recurrent unit (BiGRU), to construct a feature fusion model. This architecture effectively captures the spatiotemporal dependencies within the cardiac data. To address the challenge of optimizing the model hyperparameters, this research innovatively introduces a hybrid strategy combining Bayesian optimization with the Arctic Lemming Algorithm (ALA), which significantly enhances the model's generalization capability. The simulation results demonstrate that, the proposed intelligent hybrid optimization strategy effectively addresses the challenges of high-dimensional modeling of medical data, providing reliable intelligent decision support for the early clinical diagnosis of heart disease.

Keywords Heart disease prediction, BiTCN-BiGRU, Bayesian optimization, Arctic lemming algorithm

1 Introduction

Heart disease is one of the leading causes of death worldwide. It encompasses various types of disease, including coronary heart disease, arrhythmias, and heart failure. These conditions not only reduce the quality of life for patients, but can also lead to sudden death events. According to the World Health Organization, millions of people die annually from related diseases [1]. Early and accurate prediction is crucial for timely intervention. Traditional methods rely primarily on clinical experience and basic medical examination indicators, but exhibit subjectivity and limitations that hinder their ability to meet the demands of precision medicine. Consequently, developing efficient and accurate heart disease prediction models has become a research hotspot in both medical science and data science communities [2–4].

Current research on heart disease prediction primarily employs statistical models and machine learning methods. Traditional statistical models such as the Framingham risk

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score [5] and the SCORE model [6] establish linear risk equations through multivariate regression analysis. However, these models struggle to capture the complex nonlinear relationships between risk factors and disease, resulting in a prediction accuracy below 80%. In recent years, machine learning methods including random forest (RF) [7], support vector machine (SVM) [8], and gradient boosting tree (XGBoost) [9] have demonstrated significant potential in heart disease prediction, improving accuracy to approximately 85%. Nevertheless, these approaches exhibit limitations in handling high-dimensional feature interactions and temporal dependencies. Currently, deep learning-based methods such as convolutional neural networks (CNN) [10] and recurrent neural networks (RNN) [11] have been introduced, but most studies utilize single-network architectures that struggle to simultaneously extract spatial features and temporal dynamic characteristics, consequently maintaining accuracy around 85%. Therefore, developing a new prediction framework that can effectively integrate multi-scale features and achieve efficient hyperparameter optimization has become a current research challenge.

Recently, hybrid machine learning models have shown significant potential in the field of heart disease prediction, especially when combined with meta-heuristic optimization algorithms to improve classification performance. For instance, Niu et al. [12] proposed an improved Adaptive Curved Grey Wolf Optimization (ACGWO) algorithm, which is used to optimize the backpropagation process of neural networks and achieved an accuracy of 86.8% on the Cleveland dataset. Reference [13] proposed an Adaptive Boosting and Histogram Gradient Boosting model combined with the Sea-Horse Optimizer (SHO) and Chaos Game Optimizer (CGO), effectively improving the accuracy and reliability of heart disease prediction, with a maximum accuracy of 91.2%. In reference [14] adopted Principal Component Analysis (PCA) and Hybrid Genetic Algorithm (HGA) combined with k-means clustering, achieving a prediction accuracy of 94.06%. Yi et al. [15] combined Particle Swarm Optimization (PSO) with the Transformer model, obtaining an accuracy of 96.5% in the same task, demonstrating the potential of combining deep learning with optimization algorithms.

In recent years, hybrid prediction models integrating deep learning and optimization algorithms have demonstrated broad application prospects in the fields of finance, energy, and healthcare [16–19]. For instance, the TS-GRU model proposed in reference [16], by combining the temporal convolutional network (TCN), the gated recurrent unit (GRU) and the Sparrow Search Algorithm (SSA), significantly improves the accuracy of stock market risk prediction. Scholars et al. Literature [19] studied a hybrid model that integrates deep learning and machine learning, realizing high-precision and interpretable cancer diagnosis. Reference [20] adopted a deep learning model that combines generative adversarial networks (GAN) and BiGRU, which effectively addresses the problems of scarcity and imbalance in medical data. In reference [21], a model architecture was designed that combined a multi-scale one-dimensional convolutional network (1D-MsNet) and BiTCN, supplemented by optimization strategies. This architecture provides strong methodological support for large-scale screening of sleep apnea syndrome (SAS).

Although the hybrid model has great potentials, its performance improvement still requires efficient optimization strategies and advanced architectures for support. At the algorithmic optimization level, Bayesian optimization employs Gaussian processes

and sequential adoption strategies to achieve efficient hyperparameter optimization [22]. The Arctic Lemming Algorithm (ALA), which simulates the migratory behavior of lemmings [23], demonstrates excellent global search capabilities and rapid convergence properties, which make it suitable for complex optimization problems. At the model architecture level, the bidirectional gated recurrent unit (BiGRU) enhances context feature extraction through bidirectional gating mechanisms [24], and performs effectively in sequence prediction tasks. Meanwhile, the bidirectional temporal convolutional network (BiTCN) utilizes dilated causal convolutions to capture multiscale temporal features [25]. Its receptive field expands exponentially with increasing network depth, significantly enhancing modeling capability for long-term dependencies.

In this research, we propose a novel deep learning framework designed for hybrid intelligence to enhance the accuracy of heart disease prediction. The main contributions of this research are threefold:

- A deep integration model of BiTCN and BiGRU was designed, which can simultaneously capture the local spatiotemporal features and long-term bidirectional temporal dependencies in cardiac data.
- Combining Bayesian optimization with ALA fully leverages the efficient global exploration capability of Bayesian optimization and the powerful local search ability of ALA.
- The K-fold cross-validation, multi-index evaluation and computational efficiency analysis were adopted to ensure the reliability and reproducibility of the experimental results.

2 Dataset description

The clinical heart disease data set utilized in this work was constructed by the Cleveland Clinic Foundation, USA, on the basis of standardized clinical research protocols[26]. The data set integrates a multidimensional structured data collection comprising demographic characteristics, physiological and biochemical parameters, and medical history information. It contains a total of 303 individual samples, with each sample systematically documenting 13 clinical diagnostic characteristics and 1 categorical label. The specific feature information is detailed in Table 1.

Table 1 Description of clinical features for heart disease

Clinical features	Explanation	Range of values
Age	Patient age (year)	[29, 77]
Sex	Patient gender	0, 1
cp	History of angina pectoris	1, 2, 3, 4
trestbps	Resting blood pressure (mm/Hg)	[94, 200]
chol	Serum cholesterol content (mg/dL)	[126, 564]
fbs	Fasting blood glucose level	0, 1
restecg	Resting electrocardiogram features	0, 1, 2
thalach	Maximum heart rate (bpm)	[71, 202]
exang	Exercise induced angina	0, 1
oldpeak	ST depression relative to rest during exercise (mV)	[0, 6.2]
Slope	Slope of ST segment at peak exercise	1, 2, 3
ca	Number of large peripheral blood vessels	[0, 4]
thal	Thalassemia	0, 1, 2, 3

This data set meticulously records the clinical observation data for the patient group whose ages range from 29 to 79. In this dataset, the gender of patients is encoded using binary values, where 1 represents male and 0 represents female. The history of angina pectoris (cp) is classified into four types: 1 represents typical angina pectoris, 2 represents atypical angina pectoris, 3 represents non-cardiac pain, and 4 represents the asymptomatic type. Fasting blood glucose status (fbs) is also classified, with 0 representing fasting blood glucose below 120 mg/dL and 1 representing fasting blood glucose above 120 mg/dL. The resting electrocardiogram results (restecg) are classified into three grades: 0 represents normal electrocardiogram, 1 represents ST-T wave abnormalities, and 2 represents possible or existing left ventricular hypertrophy. The presence or absence of exercise-induced angina (exang) is also marked by binary codes, with 1 indicating pain and 0 indicating no pain. Furthermore, the slope of the ST segment during the peak exercise (slope) is classified into three grades: 1 represents an upward slope, 2 represents a flat slope, and 3 represents a downward slope. The nature of thalassemia is represented by integer coding, where 0 represents unknown, 1 represents normal, 2 represents congenital defect, and 3 represents reversible defect. Finally, the target variable (target) is used to record patient diagnosis results, with 0 indicating that there is no heart disease and 1 indicating diagnosed heart disease.

3 Heart disease prediction algorithm network architecture

The BiTCN-BiGRU model for the prediction of heart disease, optimized jointly using Bayesian and ALA approaches, is illustrated in Fig. 1. This model initially employs the Bayesian optimization algorithm to efficiently explore the parameter space for better parameter combinations. These optimized parameters subsequently serve as the initial

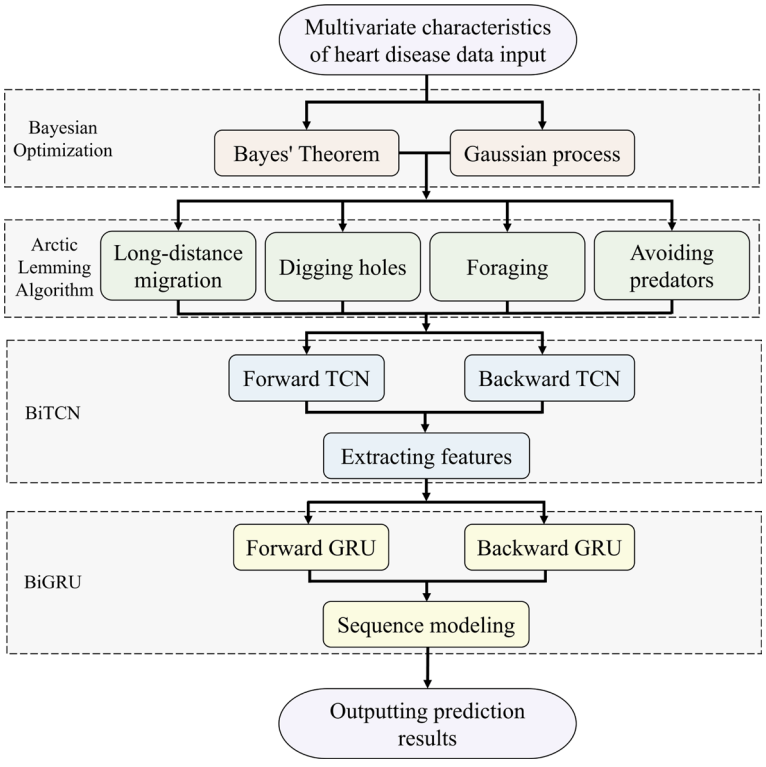


Fig. 1 Bayes-ALA-BiTCN-BiGRU model structure

population for the ALA algorithm, which continues to search the parameter space while performing adaptive adjustments to obtain an enhanced parameter set. Finally, the optimized parameters are utilized to construct the BiTCN-BiGRU model. Within this architecture, the BiTCN module performs bidirectional temporal convolution operations to extract clinical characteristics from heart disease data and analyze their correlations. The BiGRU module leverages bidirectional gated recurrent units to process cardiac data, employing its robust sequence modeling capabilities to generate more accurate prediction results.

3.1 BiTCN neural network

The BiTCN is an innovative deep learning architecture that combines the advantages of the TCN with bidirectional information flow, and excels in capturing bidirectional long-term dependencies in complex temporal data. In the task of predicting heart disease, BiTCN employs bidirectional dilated causal convolution, using bidirectional dilation factors to continuously expand the receptive field of the convolution kernel, which improves the ability to extract information and improves the quality of feature extraction, with the specific structure of this network shown in Fig. 2.

The bidirectional dilated causal convolution employed in the BiTCN network can be represented as:

$$W(t) = \sum_{k=0}^{K-1} G_F(K) \cdot S_{n-d\cdot k} + \sum_{k=0}^{K-1} G_B(K) \cdot S_{n+d\cdot k} \tag{1}$$

where G_F and G_B represent the weights of the forward and backward convolution kernel, respectively, $S_{n-d\cdot k}$ denotes the input value at the time step $n - d \cdot k$, K is the size of the convolution kernel, and d is the dilation factor.

Leveraging its unique structural design, the BiTCN effectively integrates feature information from both forward and backward temporal dimensions, significantly enhancing the model's capability to capture temporal correlations in clinical indicators for heart

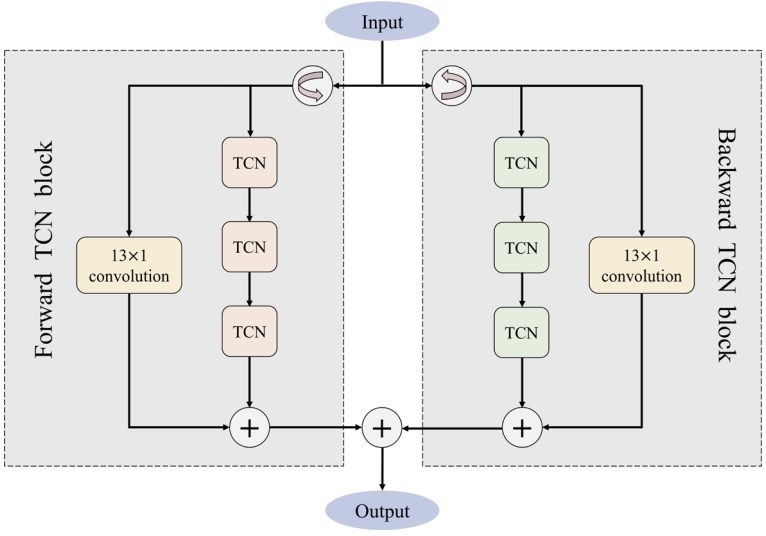


Fig. 2 BiTCN model structure

disease. The feature calculation formulas for the forward and backward TCN modules can be expressed as:

$$\overrightarrow{w_t} = \mathrm{TCN}(\overrightarrow{S_n}, \varphi, \delta, \gamma, d) \tag{2}$$

$$\overleftarrow{w_t} = \mathrm{TCN}(\overleftarrow{S_n}, \varphi, \delta, \gamma, d) \tag{3}$$

where φ denotes the dimensionality of the dilated causal convolution kernel, δ represents the parameter of the Leaky ReLU activation function, γ signifies the batch normalization rate, and d indicates the parameter for dropout regularization.

3.2 BiGRU neural network

The BiGRU is an innovative recurrent neural network architecture that comprehensively captures contextual dependencies in sequential data by integrating forward and backward information flows. In the heart disease prediction model, BiGRU is employed to further process the BiTCN output, thus enhancing the comprehensiveness, relevance, and precision of the predictions, with its specific structure illustrated in Fig. 3.

The BiGRU comprises two independent GRU layers, which simultaneously process sequential data along both forward and reverse temporal dimensions, forming complementary information flows that enhance the model's perception capability for dynamic changes in time series, with its mathematical representation expressed as follows:

$$\overrightarrow{w_n} = \mathrm{GRU}\left(S_n, \overrightarrow{w_{n-1}}\right) \tag{4}$$

$$\overleftarrow{w_n} = \mathrm{GRU}\left(S_n, \overleftarrow{w_{n-1}}\right) \tag{5}$$

$$w_n = G_f \overrightarrow{w_n} + G_b \overleftarrow{w_n} + b_n \tag{6}$$

where $\overrightarrow{w_n}$ and $\overleftarrow{w_n}$, respectively, represent the output of the forward and backward hidden layers at the time step n, S_n denotes the input data, G_f and G_b are the output weights for the hidden layers forward and backward, b_n signifies the bias term, and w_n constitutes the final output vector.

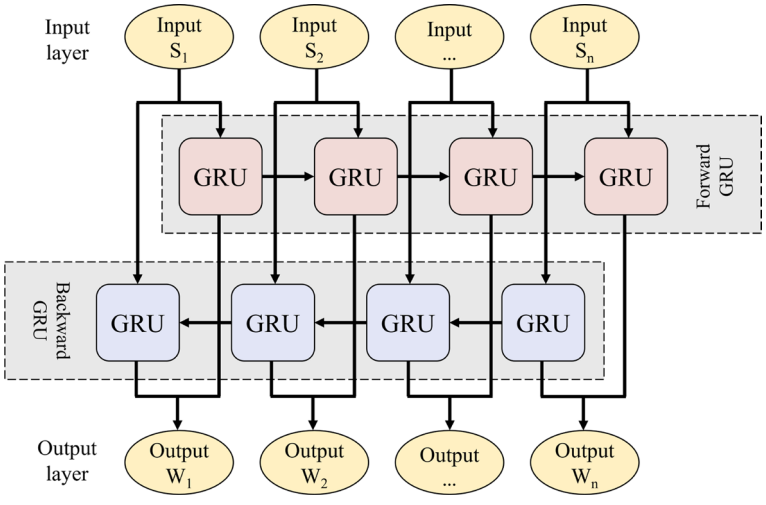


Fig. 3 BiGRU model structure

3.3 ALA optimization method in the heart disease prediction system

The ALA is an optimization algorithm that solves complex problems by simulating the exploration, adaptation, and evolution process of the lemming population. Its core innovation lies in introducing a dynamic energy factor adjustment strategy, which effectively balances the exploration (global search) and exploitation (local optimization) processes, thereby efficiently locating the global optimal solution in a complex search space. In the heart disease prediction model, the ALA algorithm further optimizes the parameters of both BiTCN and BiGRU models, balancing global exploration and local exploitation capabilities through an adaptive mechanism that dynamically adjusts the behavior of the population to avoid premature convergence and maintain population diversity, with its specific procedure illustrated in Fig. 4.

The following presents a detailed mathematical description of the iterative process of the ALA.

Step 1. Initialization:

During the algorithm initialization phase, the population matrix S_n is first randomly generated within the search space, where each decision variable $\mathrm{S}_{n,m}$ in each dimension is produced stochastically according to the following mathematical formulation:

$$\mathrm{S}_{n,m} = N_{Fm} + rand\,(N_{Bm} - N_{Fm}) \tag{7}$$

where N_{Fm} and N_{Bm} represent the upper and lower bounds of the m-th dimension, respectively, and *rand* denotes a random number between 0 and 1.

Step 2. Long-distance migration behavior:

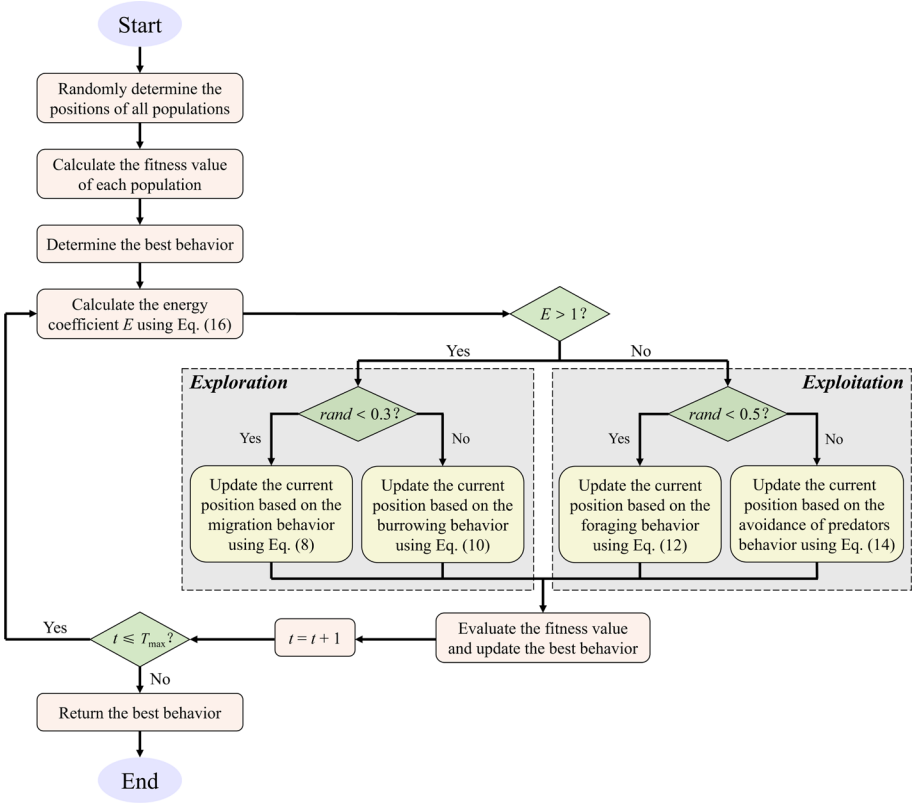


Fig. 4 Flow chart of the ALA optimization algorithm

When high population density causes food scarcity, lemmings initiate long-distance migration to seek new habitats with abundant food resources. This long-distance migration behavior can be mathematically simulated using the following formula:

$$\mathrm{S}_n(t+1) = \mathrm{S}_{best}(t) + D \times \mathrm{BM}\left[\mathrm{V}\left(\mathrm{S}_{best}(t) - \mathrm{S}_n(t)\right) + (1-\mathrm{V})\left(\mathrm{S}_n(t) - \mathrm{S}_l(t)\right)\right] \quad (8)$$

where $D \in \{-1, 1\}$ is a directional indicator that alters the search direction to help avoid local optima and enhance global search capability, V signifies a random vector uniformly distributed over the interval $[-1, 1]$ that controls inter-individual interactions and movement patterns, $\mathrm{S}_{best}(t)$ represents the current optimal solution corresponding to the known location with the most abundant food resources, $\mathrm{S}_n(t)$ indicates the current position of the *n*-th search agent, $\mathrm{S}_l(t)$ designates the position of a randomly selected search agent from the population, $\mathrm{S}_n(t+1)$ denotes the position of the *n*-th search agent at time step $t+1$, and BM characterizes a Brownian motion random vector that explores regions using dynamic and uniform step sizes, with the specific formulation of Brownian motion expressed as follows:

$$f_{BM}(x, 0, 1) = \frac{1}{\sqrt{2\pi}} \times \exp\left(-\frac{x^2}{2}\right) \quad (9)$$

Step 3. Burrowing behavior:

Lemmings randomly excavate new burrows and dig complex tunnel networks within their habitats, which not only provide safe shelters but also serve as storage spaces for food resources, with this behavior mathematically simulated by the following formulation:

$$\mathrm{S}_n(t+1) = \mathrm{S}_n(t) + D \times L \times \left(\mathrm{S}_{bet}(t) - \mathrm{S}_i(t)\right) \quad (10)$$

where $\mathrm{S}_i(t)$ and L collectively characterize the interaction mechanism between lemming individuals during new burrow excavation, with $\mathrm{S}_i(t)$ representing a randomly selected search individual from the population, and L denoting a dynamic stochastic variable related to the current iteration count, whose computational formula is expressed as follows:

$$L = rand \times \left(1 + \sin\left(\frac{t}{2}\right)\right) \quad (11)$$

Step 4. Foraging behavior:

Relying on acute olfactory and auditory senses, lemmings perform extensive random walks within their inhabited burrows according to the abundance and accessibility of food resources, with the foraging behavior being simulated using a spiral coiling mechanism:

$$\mathrm{S}_n(t+1) = \mathrm{S}_{best}(t) + D \times spiral \times rand \times \mathrm{S}_n(t) \quad (12)$$

where *spiral* denotes that the random search path during foraging exhibits a spiral structure:

$$spiral = \left\|\mathrm{S}_{bss,m}(t) - \mathrm{S}_{n,m}(t)\right\| \times \left[\sin(2\pi \times rand) + \cos(2\pi \times rand)\right] \quad (13)$$

Step 5. Predator avoidance behavior:

When encountering predator threats, lemmings utilize their high-speed mobility to escape back to burrows while implementing strategic movement trajectories to disrupt predator pursuit, with this predator avoidance behavior mathematically simulated by the following formulation:

$$\mathrm{S}_n(t+1) = \mathrm{S}_{best}(t) + D \times C \times Levy \times (\mathrm{S}_{best}(t) - \mathrm{S}_n(t)) \tag{14}$$

where C signifies the escape coefficient of lemmings, representing their escape ability which decreases with increasing iteration count, and *Levy* denotes the Lévy flight function used to simulate strategic trajectory perturbations during the escape process, with its mathematical representation expressed as:

$$Lexy = 0.01 \times \frac{u \times \sigma}{|\nu|^{\frac{1}{\beta}}}, \sigma = \left[\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\frac{\beta-1}{2}}} \right]^{\frac{1}{\beta}} \tag{15}$$

where u and ν are independent uniform random variables defined in the interval [0, 1], and β is a fixed constant with a value of 1.5.

Step 6. Balancing exploration and exploitation

To maintain a dynamic equilibrium between exploration and exploitation behaviors, an energy factor E is introduced. This factor exhibits a monotonically decreasing trend during the iterative process to regulate the behavioral decisions of lemmings. The mathematical expression for the energy factor is as follows:

$$E(t) = 4 \times \arctan\left(1 - \frac{t}{T_{\max}}\right) \times \ln\left(\frac{1}{rand}\right) \tag{16}$$

The ALA achieves efficient global optimization capabilities by modeling four typical behavior patterns of lemmings. Its energy factor mechanism dynamically adjusts the balance between exploration and exploitation, enabling the algorithm to precisely locate the global optimal solution in complex high-dimensional search spaces and demonstrate robust convergence performance.

3.4 Bayesian optimization method in the heart disease prediction system

The Bayesian optimization algorithm is a global optimization method based on sequential models. Continuously samples in the parameter space, constructs a surrogate model based on existing sampling results, and utilizes this surrogate model to predict the next most promising sampling point, thereby efficiently identifying the optimal solution of the objective function. In the heart disease prediction model, the Bayesian optimization algorithm is used to efficiently search for the better parameter combinations in the parameter space, to improve the performance of the model in the heart disease prediction task.

This algorithm builds a posterior probability distribution model of the objective function based on Bayes' theorem:

$$P(f|D) = \frac{P(D|f)P(f)}{P(D)} \tag{17}$$

where $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_t, y_t)\}$ denotes the evaluated observation data set of hyperparameters and accuracy rates, *P*(*f*) represents the prior distribution, and *P*(*D*|*f*) is the likelihood function. The algorithm simulates the behavior of the objective function by continuously updating the posterior distribution.

The algorithm employs a Gaussian process (GP) as the surrogate model:

$$f(x) \sim \mathcal{G}P\left(m(x), k(x, x')\right) \tag{18}$$

where the mean function $m(x)$ is typically set to zero, and the Matérn kernel function is adopted for the covariance function , $k(x, x')$:

$$k_{\nu-5/2}(r) = \sigma^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2\right) \exp\left(-\sqrt{5}r\right), r = \sqrt{\sum_d (x_d - x_d')^2 / \theta_d^2} \tag{19}$$

In each iteration, the Bayesian optimization algorithm computes the value of the acquisition function for each sampling point based on the current surrogate model. The expected improvement (EI) function is used to quantify the value of sampling at a specific point, balancing the trade-off between exploration and exploitation. The formulation of the EI function is given as follows:

$$EI(x) = \left[\mu(x) - f(x^+) - \xi\right] \Phi(Z) + \sigma(x)\phi(Z) \tag{20}$$

where $\mu(x)$ and $\sigma(x)$ denote the mean and standard deviation, respectively, of the Gaussian process at the point x, $f(x^+)$ represents the optimal value currently known of the objective function, ξ is a hyperparameter that balances exploration and exploitation, and $\Phi(Z)$ and $\Phi(Z)$ correspond to the cumulative distribution function and probability density function, respectively, of the standard normal distribution.

The Bayesian optimization algorithm selects the point with the highest value of the acquisition function as the next sampling point, and evaluates the objective function at this location. Subsequently, this new sample point is incorporated into the existing data set, and the surrogate model is updated:

$$x_{n+1} = \arg\max_{x \in X} \mathrm{EI}(x) \tag{21}$$

4 Simulation results and evaluation

This section presents a simulation study investigating the performance of Bayesian optimization and ALA optimization on the base model (BiTCN-BiGRU) for clinical heart disease prediction tasks. The experiment utilized a heart disease dataset comprising 13 clinical features. A standardized preprocessing pipeline was implemented, which included handling missing values, followed by feature encoding, and finally feature scaling and normalization. After preprocessing, the data set is partitioned into training and test sets in an 8:2 ratio, with a stratified sampling strategy that ensures consistent class distribution between training and test sets, thereby ensuring the reliability of model training. During the model evaluation phase, k-fold cross-validation (k=3) was introduced to further verify the generalization capability and stability of the model.The specific parameter settings are as described in Table 2.

Fig. 5 presents the ranking results of 13 clinical features based on the Gini index method embedded in the random forest model. This bar chart quantifies the relative

Table 2 The search range of hyperparameters for the Bayes-ALA joint optimization algorithm

Parameter name	Parameter description	Set the range
tcn_filters	Number of convolutional filters	[16, 64]
tcn_kernel_size	Convolution kernel size	[2, 4]
gru_units	Number of hidden units in GRU	[16, 64]
dropout_rate	Dropout rate	[0.3, 0.6]
learning_rate	Learning rate	[0.00001, 0.0005]
dense_units	Number of units in the fully connected layer	[16, 64]
epochs	Number of training rounds	60
batch_size	Batch size	32

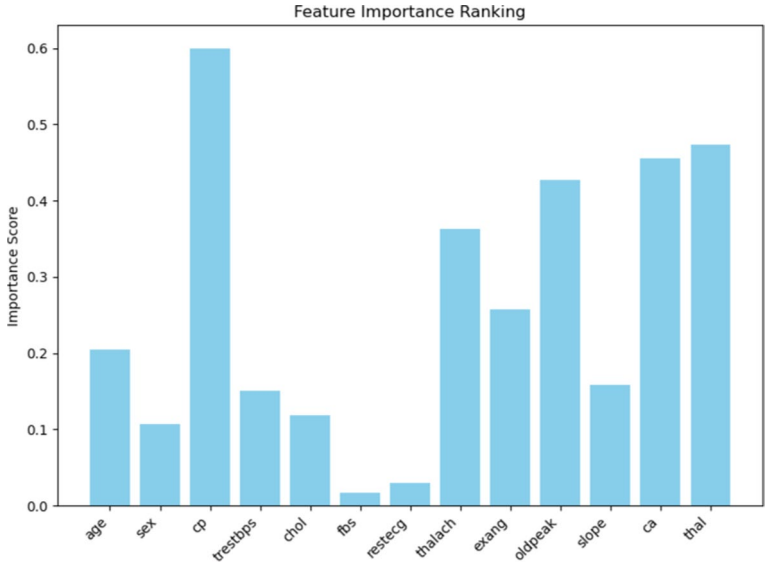
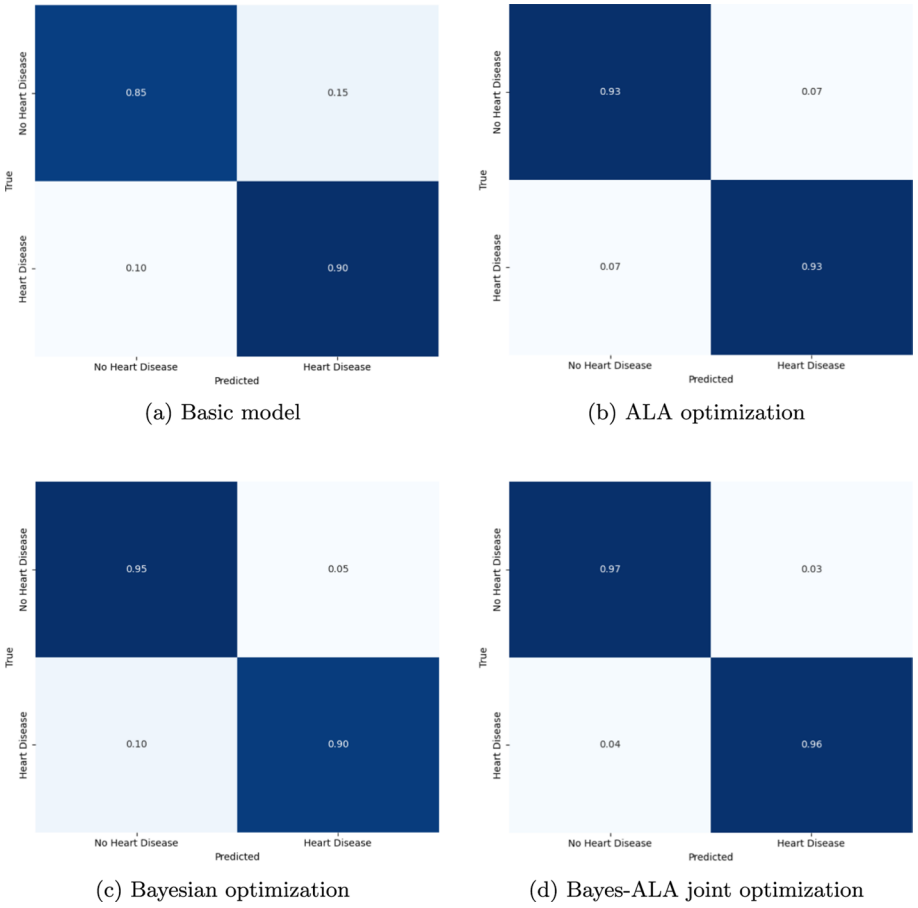


Fig. 5 Feature importance

contribution of each feature to improving the accuracy of the heart disease prediction model. The results of the analysis show that features such as cp, thalach, and oldpeak have a relatively high degree of contribution, which exert a decisive influence on the prediction performance of the heart disease model. Other features, including age, trestbps, and chol, although having a relatively lower degree of contribution, still play a synergistic role in the model to collectively improve the prediction accuracy. This feature importance analysis not only provides key empirical evidence for subsequent optimization of the model but also offers intuitive decision support references for clinicians, thus facilitating diagnostic decision making.

Fig. 6 presents the confusion matrices of various prediction models, where the percentage data reflect the discriminative capabilities of the models for healthy individuals (No Heart Disease) and patients (Heart Disease). The results demonstrate that the baseline model achieves recognition accuracies of 85% and 90% for healthy individuals and patients, respectively, while exhibiting high missed diagnosis and misdiagnosis rates of 15% and 10%. Both the ALA-optimized model and the Bayesian-optimized model show reductions in missed diagnosis and misdiagnosis rates compared to the baseline model, yet each exhibits certain limitations. In contrast, the proposed hybrid Bayes-ALA optimization model elevates the recognition rates for patients and healthy individuals to 96% and 97%, respectively, thus further optimizing the accuracy of the prediction.



(a) Basic model (b) ALA optimization

(c) Bayesian optimization (d) Bayes-ALA joint optimization

Fig. 6 Confusion matrix of different algorithm prediction models

Table 3 Comparison of the performance of different prediction models

Model	Accuracy	Recall	F1 score	AUC	Training time (s)
Base (BiTCN-biGRU)	0.873	0.895	0.879	0.940	15.641
ALA Optimized	0.931	0.933	0.933	0.970	12.623
Bayes Optimized	0.927	0.905	0.927	0.969	12.582
Bayes+ALA Optimized	0.966	0.962	0.967	0.986	16.243

To systematically evaluate the generalization ability and stability of the optimized model proposed in this research, based on the confusion matrix results of the four prediction models, the accuracy, recall rate, F1 score, AUC value, and model training time of each prediction model were further calculated, as shown in Table 3.

According to the comparison results in Table 3, it can be seen that the model optimized by the Bayes-ALA joint optimization shows significant performance advantages in the task of heart disease prediction. This model outperforms the baseline model in terms of precision, recall, F1 score, and AUC value. Although its training time is slightly increased, in the medical disease prediction scenario, the priority of model accuracy for clinical decision making is higher than that of efficiency considerations.

Fig. 7 compares the receiver operating characteristic (ROC) curves of various prediction models, accompanied by their corresponding area under the curve (AUC) and accuracy (Acc) values. The ROC curve illustrates the relationship between the true positive

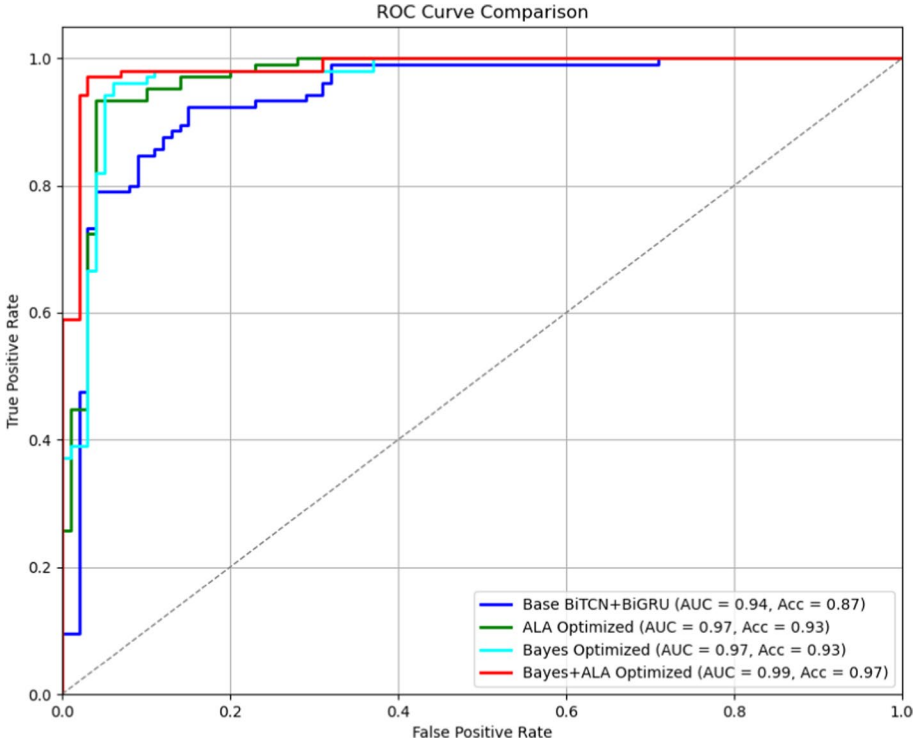


Fig. 7 ROC curve and prediction accuracy

rate and false positive rate across different classification thresholds, while the AUC quantifies the model's overall classification performance. A higher AUC value indicates a stronger capability of the model to distinguish between positive and negative samples. The results demonstrate that the proposed hybrid Bayes-ALA optimization model achieves an exceptional AUC value of 99% and a prediction accuracy of 97%, surpassing the performance of models optimized solely with either Bayesian optimization or ALA applied to the baseline model. This hybrid approach exhibits significant improvements in both discriminative capability and predictive precision.

5 Conclusion

This paper investigates a BiTCN-BiGRU model for heart disease prediction based on the joint optimization of Bayesian and ALA. The model couples the long-term feature extraction capability of BiTCN with the sequence modeling ability of BiGRU, effectively capturing the complex patterns of multi-feature characteristics in heart disease data. To further optimize model performance, this study innovatively introduces a Bayesian-ALA joint optimization method for hyperparameter tuning. Through experimental comparison, the efficiency of parameter optimization in deep learning models using the bayes-ALA joint optimization method has been verified, providing a new technical path for medical auxiliary diagnosis. However, the current dataset size used in this study is still relatively limited. In future work, efforts will be made to conduct further verification on larger-scale or external independent datasets to enhance the generalization ability and clinical applicability of the model.

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Author contributions
The specific contributions of all authors are as follows: [Wenhui Jiang]: Responsible for concept research and algorithm design, execution of major experiments, and writing of the original manuscript. [Lexiang Zheng.]: Responsible for algorithm design and data analysis. [Yue Tian.]: Responsible for theoretical analysis, manuscript review and revision, and supervision. [Yau Hee Kho]: Responsible for reviewing manuscripts, conducting critical revisions, and supervision.

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Data availability
This research used public database data. Data acquisition website: https://archive.ics.uci.edu/ml/datasets/Heart+Disease.deer

Code availability
Codes can be shared on request.

Declarations

Ethics approval and consent to participate
Not applicable (retrospective anonymized data used).

Consent to publication
Not applicable.

Competing interests
The authors declare no competing interests.

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Not applicable.

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References

1. Mensah GA, Fuster V, Murray CJL. Global burden of cardiovascular diseases and risks, 1990–2022. J Am Coll Cardiol. 2023;82(25):2350–473.
2. Hassan CAU, Iqbal J, Irfan R. Effectively predicting the presence of coronary heart disease using machine learning classifiers. Sensors. 2022;22(19):7227.
3. Zhou C, Dai P, Hou A. A comprehensive review of deep learning-based models for heart disease prediction. Artif Intell Rev. 2024;57(10):263.
4. Cai Y, Cai YQ, Tang LY. Artificial intelligence in the risk prediction models of cardiovascular disease and development of an independent validation screening tool: a systematic review. BMC Med. 2024;22(1):56.
5. Artigao-Rodenas LM, Carbayo-Herencia JA, Divisón-Garrote JA. Framingham risk score for prediction of cardiovascular diseases: a population-based study from southern Europe. PLoS One. 2013;8(9):73529.
6. Arvanitaki A, Ntiloudi D, Giannakoulas G. Prediction models and scores in adult congenital heart disease. Curr Pharm Des. 2021;27(10):1232–44.
7. Wang J, Rao C, Goh M. Risk assessment of coronary heart disease based on cloud-random forest. Artif Intell Rev. 2023;56(1):203–32.
8. Mondal A, Mondal B, Chakraborty A. Heart disease prediction using support vector machine and artificial neural network. Artif Intell Applications. 2024;2(1):45–51.
9. Budholiya K, Shrivastava SK, Sharma V. An optimized xgboost based diagnostic system for effective prediction of heart disease. J King Saud Univ. 2022;34(7):4514–23.
10. Dutta A, Batabyal T, Basu M. An efficient convolutional neural network for coronary heart disease prediction. Expert Syst Appl. 2020;159:113408.
11. Choi E, Schuetz A, Stewart WF. Using recurrent neural network models for early detection of heart failure onset. J Am Med Inform Assoc. 2017;24(2):361–70.
12. Zhang H, Li C, Tian X, et al. Enhancing heart disease prediction accuracy with hybrid machine learning. Comput Methods Biomech Biomed Engin. 2025; 1–12.
13. Islam MT, Rafa SR, Kibria, MG. Early prediction of heart disease using PCA and hybrid genetic algorithm with k-means. 2020 23rd Intl Conf Comput Info Technol (ICCIT). IEEE, 2020; 1–6 .
14. Yi J, Yu P, Huang T, et al. Optimization of transformer heart disease prediction model based on particle swarm optimization algorithm. 2024 6th Intl Conf on Frontier Tech Info Comput (ICFTIC). IEEE, 2024; 1109–1113.
15. Niu S, Zhou Y, Li Z, et al. An improved grey wolf optimization algorithm for heart disease prediction. IEEE Intl Conf on Advanced Infocomm Tech. Springer Nature Singapore, 2023; 527–42.
16. Zhang Y, Heinz DF. Ts-gru: a stock gated recurrent unit model driven via neuro-inspired computation. Electronics. 2024;13(23):4659.

17. Fu H, Zhang J, Xie S. A novel improved variational mode decomposition-temporal convolutional network-gated recurrent unit with multi-head attention mechanism for enhanced photovoltaic power forecasting. Electronics. 2024;13(10):1837.
18. Junaid HHS, Daneshfar F, Mohammad MA. Automatic colorectal cancer detection using machine learning and deep learning based on feature selection in histopathological images. Biomed Signal Process Control. 2025;107:107866.
19. Hosseini E, Al-Ghaili AM, Kadir DH, et al. The evolutionary convergent algorithm: a guiding path of neural network advancement. IEEE Access. 2024. https://doi.org/10.1109/ACCESS.2024.3452511.
20. Fatima T, Yang W, Xia, K, et al. Deep learning approach combining GAN and BiGRU for diabetes prediction. 2024 3rd Intl Conf Robot Artif Intell Intelligent Ctrl (RAIIC). IEEE, 2024; 346–349.
21. Zou L, Liu G. Multiscale bidirectional temporal convolutional network for sleep apnea detection based on wearable photoplethysmography bracelet. IEEE J Biomed Health Inform. 2023;28(3):1331–40.
22. Wu J, Chen XY, Zhang H. Hyperparameter optimization for machine learning models based on bayesian optimization. J Elec Sci Tech. 2019;17(1):26–40.
23. Xiao Y, Cui H, Khurma RA. Artificial lemming algorithm: a novel bionic meta-heuristic technique for solving real-world engineering optimization problems. Artif Intell Rev. 2025;58(3):84.
24. Jia C, Tian Y, Shi Y. State of health prediction of lithium-ion batteries based on bidirectional gated recurrent unit and transformer. Energy. 2023;285:129401.
25. Zhang D, Chen B, Zhu H. Short-term wind power prediction based on two-layer decomposition and BITCN-BiLSTM-attention model. Energy. 2023;285:128762.
26. Janosi A, Steinbrunn W, Pfisterer M, Detrano R. Heart Disease Dataset. UCI Machine Learning Repository https://archive.ics.uci.edu/ml/datasets/Heart+Disease, accessed on Sept. 02, 2019 (1988).